# Functions & Their Graphs

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Review: Relations and functions**

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the $xy$-plane with vertex $(2, 9)$ and $y$-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of $g$ passes through the vertex of the graph of $f$, what is the other $x$-coordinate where the graphs of $f$ and $g$ intersect?

We solve the problem in three steps: 1. Determine the quadratic function $f(x)$: The vertex is given as $(h, k) = (2, 9)$, so the function has the form: $f(x) = a(x - 2)^2 + 9$ Since the graph passes through the point $(0, 5)$: $5 = a(0 - 2)^2 + 9 \Rightarrow 4a + 9 = 5 \Rightarrow 4a = -4 \Rightarrow a = -1$ Expanding $f(x)$: $f(x) = -(x - 2)^2 + 9 = -(x^2 - 4x + 4) + 9 = -x^2 + 4x + 5$ 2. Determine the linear function $g(x)$: The line $g(x) = mx + 1$ passes through the vertex $(2, 9)$.

**2** **[Foundational]** • **Target: Function families and transformations**

The graph of the polynomial function $f(x) = x^3 + ax^2 + bx - 6$ has a local maximum at $x = -3$ and a local minimum at $x = 1$. Find the values of $a$ and $b$.

For a polynomial function to have local extrema at specific x-values, its derivative must be zero at those points. The derivative of $f(x) = x^3 + ax^2 + bx - 6$ is $f'(x) = 3x^2 + 2ax + b$. Since there are local extrema at $x = -3$ and $x = 1$, we have $f'(-3) = 0$ and $f'(1) = 0$. Substituting $x = -3$ into $f'(x)$: $3(-3)^2 + 2a(-3) + b = 0$ $3(9) - 6a + b = 0$ $27 - 6a + b = 0$ (Equation 1) Substituting $x = 1$ into $f'(x)$: $3(1)^2 + 2a(1) + b = 0$ $3 + 2a + b = 0$ (Equation 2) Now we have a system of two line.

**3** **[Foundational]** • **Target: Choosing appropriate models**

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an $x$-intercept at $(3, 0)$. A second function $g$ is defined by $g(x) = f(x - 2) + k$, where $k$ is a constant. If the graph of $g$ passes through the origin $(0, 0)$, what is the value of $k$?

Step 1: Write the equation of $f(x)$ in vertex form. Since the vertex is $(h, k_v) = (1, 8)$, the quadratic function has the form: $f(x) = a(x - 1)^2 + 8$ Step 2: Use the known point $(3, 0)$ to solve for $a$. $f(3) = a(3 - 1)^2 + 8 = 0 \Rightarrow 4a + 8 = 0 \Rightarrow a = -2$ Thus, $f(x) = -2(x - 1)^2 + 8$. Step 3: Apply the condition that $g(x)$ passes through $(0, 0)$. $g(0) = f(0 - 2) + k = 0 \Rightarrow f(-2) + k = 0$ Step 4: Evaluate $f(-2)$. $f(-2) = -2(-2 - 1)^2 + 8 = -2(-3)^2 + 8 = -2(9) + 8 = -10$ Step.

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**4** [Foundational] • Target: Domain and range and continuity

What is the horizontal asymptote of the graph of the function $f(x) = 3 \cdot 2^{x+1} - 5$?

An exponential function in the standard form $f(x) = a \cdot b^{x-h} + k$ (with base $b > 1$) has a horizontal asymptote determined by its vertical shift, $y = k$. As $x \to -\infty$, the exponential power $2^{x+1} \to 0$. Substituting this into the function yields: $\lim_{x \to -\infty} f(x) = 3(0) - 5 = -5$ Thus, the horizontal asymptote is the horizontal line $y = -5$. Common Misconceptions: - $y = 0$: Assumes the asymptote is always the $x$-axis, forgetting the vertical shift by $-5$..

**5** [Foundational] • Target: Characteristics from a graph

The quadratic function $f(x) = ax^2 + bx + c$ has its vertex at $(-1, 5)$. If the graph of $f(x)$ also passes through the point $(2, -13)$, find the value of $b$.

The vertex form of a quadratic function is $f(x) = a(x-h)^2 + k$, where $(h, k)$ is the vertex. Given the vertex is $(-1, 5)$, we have $f(x) = a(x-(-1))^2 + 5 = a(x+1)^2 + 5$. The graph passes through the point $(2, -13)$, so we can substitute these values into the equation: $-13 = a(2+1)^2 + 5$. This simplifies to $-13 = a(3)^2 + 5$, so $-13 = 9a + 5$. Subtracting 5 from both sides gives $-18 = 9a$, which means $a = -2$. Now, we expand the vertex form to find the standard form $f(x) = ax^2 + bx + c$.

**6** [Foundational] • Target: Average rates of change

The graph of the polynomial function $f(x) = x^3 - ax^2 + bx - 6$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. Determine the value of $a + b$.

To find the local extrema of a polynomial function, we need to find its critical points by taking the derivative and setting it to zero. The derivative of $f(x)$ is $f'(x) = 3x^2 - 2ax + b$. Since $f(x)$ has local extrema at $x = -1$ and $x = 3$, these are the roots of $f'(x) = 0$. Thus, we can write $f'(x)$ in factored form: $f'(x) = k(x-(-1))(x-3) = k(x+1)(x-3)$ for some constant $k$. Expanding this, we get $f'(x) = k(x^2 - 2x - 3)$. Comparing the coefficient of $x^2$ in $f'(x) = 3x^2 - 2ax + b$ and.

**7** [Foundational] • Target: Review: Relations and functions

Consider the function $f(x) = ax^2 + bx + c$. The graph of $y = f(x)$ is a parabola that passes through the points $(-1, 5)$ and $(2, -7)$. If the vertex of the parabola is located at $x = -1/2$, find the value of $c$.

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ occurs at $x = -b/(2a)$. We are given that the vertex occurs at $x = -1/2$, so $-b/(2a) = -1/2$, which implies $b = a$. The parabola passes through $(-1, 5)$, so $f(-1) = a(-1)^2 + b(-1) + c = a - b + c = 5$. Since $b = a$, this becomes $a - a + c = 5$, which simplifies to $c = 5$. The parabola also passes through $(2, -7)$, so $f(2) = a(2)^2 + b(2) + c = 4a + 2b + c = -7$. Substituting $b = a$ and $c = 5$ into this equation gives $4a + 2a + 5 = -7$, so.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Function families and transformations

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of $a$ and the value of $b$.

Let the function be $f(x) = a \cdot b^x$. We are given that the graph passes through $(1, 12)$ and $(3, 108)$. Substituting these points into the function equation, we get two equations: 1) $a \cdot b^1 = 12 \Rightarrow ab = 12$ 2) $a \cdot b^3 = 108$ From equation (1), we can express $a$ in terms of $b$: $a = \frac{12}{b}$. Substitute this expression for $a$ into equation (2): $\left(\frac{12}{b}\right) \cdot b^3 = 108$ $12 \cdot b^2 = 108$ $b^2 = \frac{108}{12}$ $b^2 = 9$ Since $b$ must be positive for an exponen.

**9** [Intermediate] • Target: Choosing appropriate models

Consider the function $f(x) = ax^2 + bx - 5$. If the graph of $f(x)$ passes through the points $(-2, 3)$ and $(1, -3)$, find the value of $a - b$.

The function is given by $f(x) = ax^2 + bx - 5$. Since the graph passes through $(-2, 3)$, we have $f(-2) = 3$: a(-2)^2 + b(-2) - 5 = 3 \\ 4a - 2b - 5 = 3 \\ 4a - 2b = 8 \\ 2a - b = 4 \quad (1) Since the graph passes through $(1, -3)$, we have $f(1) = -3$: a(1)^2 + b(1) - 5 = -3 \\ a + b - 5 = -3 \\ a + b = 2 \quad (2) Now we have a system of two linear equations with two variables: 1) $2a - b = 4$ 2) $a + b = 2$ Adding equation (1) and equation (2): (2a - b) + (a + b) = 4 + 2 \\ 3a = 6 \\ a = 2 Substituting.

**10** [Intermediate] • Target: Domain and range and continuity

The graph of the sinusoidal function $y = A\sin(Bx + C) + D$ is shown below. The function has a maximum value of 5 and a minimum value of -1. The graph passes through the point $(0, 2)$ and has a period of $\frac{\pi}{2}$. Find the value of $A$ and $B$. Assume $A > 0$ and $B > 0$.

The amplitude $A$ is half the difference between the maximum and minimum values: $A = \frac{5 - (-1)}{2} = \frac{6}{2} = 3$. Since $A > 0$ is assumed, $A = 3$. The vertical shift $D$ is the average of the maximum and minimum values: $D = \frac{5 + (-1)}{2} = \frac{4}{2} = 2$. The period of the function is given by $T = \frac{2\pi}{|B|}$. We are given that the period is $\frac{\pi}{2}$, so $\frac{2\pi}{B} = \frac{\pi}{2}$ (since $B > 0$). Solving for $B$, we get $B = \frac{2\pi}{\pi/2} = 4$. Thus, $A = 3$ and $B = 4$.

**11** [Intermediate] • Target: Characteristics from a graph

The graph of the function $f(x) = a\sqrt{x - h} + k$ is shown below. The function passes through the points $(5, 10)$ and $(13, 16)$. Determine the value of $a$ and the coordinates of the vertex $(h, k)$.

The vertex of the graph is $(h, k)$. From the graph, we can infer that the vertex is at $(1, 4)$. Thus, $h = 1$ and $k = 4$. The function can be written as $f(x) = a\sqrt{x - 1} + 4$. We are given that the graph passes through the point $(5, 10)$. Substituting these values into the function: $10 = a\sqrt{5 - 1} + 4$. This simplifies to $10 = a\sqrt{4} + 4$, so $10 = 2a + 4$. Subtracting 4 from both sides gives $6 = 2a$, which means $a = 3$. Therefore, $a = 3$ and the vertex is $(h, k) = (1, 4)$.

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**12** **[Intermediate]** • **Target: Average rates of change**

For the rational function $f(x) = \frac{x-2}{x+3}$, what value of $x$ is excluded from the domain?

A rational function is undefined when its denominator is equal to zero. For the function $f(x) = \frac{x-2}{x+3}$, the denominator is $x + 3$. To find the excluded value from the domain, we set the denominator equal to zero and solve for $x$: $x + 3 = 0$. Subtracting 3 from both sides gives $x = -3$. Therefore, $x = -3$ is excluded from the domain.

**13** **[Intermediate]** • **Target: Review: Relations and functions**

Consider the rational function $f(x) = \frac{x^2 - 5x + 6}{x^2 - 7x + 12}$. Let $g(x)$ be the function obtained by simplifying $f(x)$. Determine the value of $g(x)$ as $x \to \infty$.

The function is given by $f(x) = \frac{x^2 - 5x + 6}{x^2 - 7x + 12}$. We first factor the numerator and the denominator: Numerator: $x^2 - 5x + 6 = (x - 2)(x - 3)$ Denominator: $x^2 - 7x + 12 = (x - 3)(x - 4)$ So, $f(x) = \frac{(x-2)(x-3)}{(x-3)(x-4)}$. For $x \neq 3$, we can simplify this to $g(x) = \frac{x-2}{x-4}$. We need to find the limit of $g(x)$ as $x \to \infty$. We can do this by dividing both the numerator and the denominator by the highest power of $x$ in the denominator, which is $x$.

**14** **[Intermediate]** • **Target: Function families and transformations**

The bar graph shows the distribution of the number of goals scored per game by a soccer team across all 40 games of its regular season. If the team plays $k$ post-season games and scores exactly 3 goals in each of those games, the mean number of goals scored per game across all regular-season and post-season games combined will be exactly 2. What is the value of $k$?

To find $k$, first determine the total number of goals scored during the regular season from the bar graph: - Games with 0 goals: $8 \Rightarrow 0 \times 8 = 0$ goals - Games with 1 goal: $14 \Rightarrow 1 \times 14 = 14$ goals - Games with 2 goals: $12 \Rightarrow 2 \times 12 = 24$ goals - Games with 3 goals: $6 \Rightarrow 3 \times 6 = 18$ goals Total regular-season games: $8 + 14 + 12 + 6 = 40$. Total regular-season goals: $0 + 14 + 24 + 18 = 56$. When the team plays $k$ additional games scoring 3 goals in each.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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**15** [Challenging] • Target: Choosing appropriate models • Cross-Lesson: 4.01 Rational Expressions

The function $f(x) = a \cdot 2^x + k$ has a horizontal asymptote at $y = 3$ and passes through the point $(0, 7)$, where $a$ and $k$ are real constants. What is the value of $f(2)$?

For an exponential function of the form $f(x) = a \cdot 2^x + k$, as $x \to -\infty$, $2^x \to 0$, which gives the horizontal asymptote $y = k$. Since the horizontal asymptote is given as $y = 3$, we have: $k = 3$ Next, the graph passes through $(0, 7)$, so $f(0) = 7$: $a \cdot 2^0 + 3 = 7$ Since $2^0 = 1$, this simplifies to: $a + 3 = 7 \Rightarrow a = 4$ Thus, the function is $f(x) = 4 \cdot 2^x + 3$. Evaluating at $x = 2$: $f(2) = 4 \cdot 2^2 + 3 = 4 \cdot 4 + 3 = 16 + 3 = 19$ Common errors: - Se.

**16** [Challenging] • Target: Domain and range and continuity • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

What is the period of the trigonometric function $f(x) = 5\cos(4x) - 2$?

For a sinusoidal function written in the standard form $f(x) = A\cos(Bx - C) + D$, the period $T$ is determined by the coefficient of $x$: $T = \frac{2\pi}{|B|}$ For the given function $f(x) = 5\cos(4x) - 2$, the frequency coefficient is $B = 4$. Substituting this into the period formula gives: $T = \frac{2\pi}{4} = \frac{\pi}{2}$ Misconception analysis: - $\frac{\pi}{4}$ results from incorrectly applying the period formula for the tangent function, $\frac{\pi}{|B|}$. - $2\pi$ is the period of the parent function.

**17** [Challenging] • Target: Characteristics from a graph • Cross-Lesson: 6.01 Radicals & Rational Exponents

Consider the quadratic function $f(x) = x^2 - 6x + 8$. What are the roots of this function?

To find the roots of the quadratic function $f(x) = x^2 - 6x + 8$, we set $f(x) = 0$ and solve for $x$. This gives us the equation $x^2 - 6x + 8 = 0$. We can factor this quadratic equation. We are looking for two numbers that multiply to 8 and add up to -6. These numbers are -2 and -4. Therefore, the factored form of the equation is $(x - 2)(x - 4) = 0$. Setting each factor to zero, we get $x - 2 = 0$ or $x - 4 = 0$. Solving these equations, we find $x = 2$ and $x = 4$. These are the roots of the function.

**18** [Challenging] • Target: Average rates of change • Cross-Lesson: 7.01 Sequences & Series

Determine all continuous even functions $f$ satisfying $f(f(x)) = x$.

If $f$ is even and involutory: $f(x) = \pm x$ work ($f(-x) = f(x)$ satisfied by $|x|$... but $f(f(x)) = ||x|| = |x| eq x$ for $x < 0$). For $f(x) = x$: even? Only if $x = -x$, so $f(x) = -x$ with $f(-x) = -(-x) = x eq -x$ generally. Actually $f(x) = x$ is both even and odd only if $f = 0$. Simplest: $f(x) = x$ (for $x = 0$ even trivially). $f(x) = -x$ is odd not even. Best answer: $f(x) = x$.

**19** [Challenging] • Target: Review: Relations and functions • Cross-Lesson: 8.01 Conic Sections

What is the value of the exponential function $f(x) = 2^x$ when $x = 3$?

The exponential function is given by $f(x) = 2^x$. To find the value of the function when $x = 3$, we substitute 3 for $x$ into the function's formula: $f(3) = 2^3$ $2^3 = 2 \times 2 \times 2 = 8$ Therefore, the value of the function $f(x) = 2^x$ when $x = 3$ is 8.

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**20** [Challenging] • Target: Function families and transformations • Cross-Lesson: 9.01 Trigonometric Functions

Consider the polynomial function $P(x) = 2x^3 - 5x^2 + 3x - 7$. What is the value of $P(2)$?

To find the value of $P(2)$, substitute $x = 2$ into the polynomial function: $P(2) = 2(2)^3 - 5(2)^2 + 3(2) - 7$ $P(2) = 2(8) - 5(4) + 6 - 7$ $P(2) = 16 - 20 + 6 - 7$ $P(2) = -4 + 6 - 7$ $P(2) = 2 - 7$ $P(2) = -5$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Review: Relations and functions

For the model $f(x) = 2x + 3$ in review: relations and functions, evaluate $f(2)$.

**a.** 7 ✓

**b.** 9

**c.** 4

**d.** 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

**22** [Foundational] • Target: Review: Relations and functions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

**a.** $x = 2$ and $x = 5$

**b.** $x = 2$ **and** $x = 4$ ✓

**c.** $x = -2$ and $x = -4$

**d.** $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

**23** [Foundational] • Target: Review: Relations and functions

For the model $f(x) = 2x + 3$ in review: relations and functions, evaluate $f(5)$.

**a.** 13 ✓

**b.** 15

**c.** 10

**d.** 17

Substitute $x = 5$: $f(5) = 2(5) + 3 = 13$.

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**24** [Foundational] • Target: Function families and transformations

For the model $f(x) = 3x + 4$ in function families and transformations, evaluate $f(2)$.

**a.** 10 ✓

**b.** 13

**c.** 6

**d.** 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

**25** [Foundational] • Target: Function families and transformations

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

**a.** $x = 2$ and $x = 6$

**b.** $x = 2$ **and** $x = 5$ ✓

**c.** $x = -2$ and $x = -5$

**d.** $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

**26** [Foundational] • Target: Function families and transformations

For the model $f(x) = 3x + 4$ in function families and transformations, evaluate $f(5)$.

**a.** 19 ✓

**b.** 22

**c.** 15

**d.** 23

Substitute $x = 5$: $f(5) = 3(5) + 4 = 19$.

**27** [Foundational] • Target: Choosing appropriate models

For the model $f(x) = 4x + 5$ in choosing appropriate models, evaluate $f(2)$.

**a.** 13 ✓

**b.** 17

**c.** 8

**d.** 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

**28** [Foundational] • Target: Choosing appropriate models

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

**a.** $x = 2$ and $x = 7$

**b.** $x = 2$ **and** $x = 6$ ✓

**c.** $x = -2$ and $x = -6$

**d.** $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** **[Intermediate]** • **Target: Review: Relations and functions**

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

**a.** $x = 1$ and $x = 4$

**b.** $x = 1$ **and** $x = 3$ ✓

**c.** $x = -1$ and $x = -3$

**d.** $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

**30** **[Intermediate]** • **Target: Review: Relations and functions**

A data set recorded for review: relations and functions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

**a.** 14.5

**b.** 15.5

**c.** 16.5 ✓

**d.** 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives $16.5$.

**31** **[Intermediate]** • **Target: Review: Relations and functions**

For the model $f(x) = 2x + 3$ in review: relations and functions, evaluate $f(4)$.

**a.** 11 ✓

**b.** 13

**c.** 8

**d.** 15

Substitute $x = 4$: $f(4) = 2(4) + 3 = 11$.

**32** **[Intermediate]** • **Target: Review: Relations and functions**

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

**a.** $x = 4$ and $x = 7$

**b.** $x = 4$ **and** $x = 6$ ✓

**c.** $x = -4$ and $x = -6$

**d.** $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

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**33** [Intermediate] • Target: Function families and transformations

A data set recorded for function families and transformations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** 15.5

**b.** 16.5

**c.** 17.5 ✓

**d.** 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by $4$ gives $17.5$.

**34** [Intermediate] • Target: Function families and transformations

For the model $f(x) = 3x + 4$ in function families and transformations, evaluate $f(3)$.

**a.** 13 ✓

**b.** 16

**c.** 9

**d.** 17

Substitute $x = 3$: $f(3) = 3(3) + 4 = 13$.

**35** [Intermediate] • Target: Function families and transformations

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

**a.** $x = 3$ and $x = 7$

**b.** $x = 3$ **and** $x = 6$ ✓

**c.** $x = -3$ and $x = -6$

**d.** $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

**36** [Intermediate] • Target: Choosing appropriate models

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

**a.** $x = 1$ and $x = 6$

**b.** $x = 1$ **and** $x = 5$ ✓

**c.** $x = -1$ and $x = -5$

**d.** $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

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**37** [Intermediate] • Target: Choosing appropriate models

For the model $f(x) = 4x + 5$ in choosing appropriate models, evaluate $f(3)$.

**a.** 17 ✓

**b.** 21

**c.** 12

**d.** 21

Substitute $x = 3$: $f(3) = 4(3) + 5 = 17$.

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Choosing appropriate models • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 4x + 5$ in choosing appropriate models, evaluate $f(4)$.

**a.** 21 ✓

**b.** 25

**c.** 16

**d.** 25

Substitute $x = 4$: $f(4) = 4(4) + 5 = 21$.

**39** [Challenging] • Target: Choosing appropriate models • Cross-Lesson: 8.01 Conic Sections

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

**a.** $x = 4$ and $x = 9$

**b.** $x = 4$ **and** $x = 8$ ✓

**c.** $x = -4$ and $x = -8$

**d.** $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

**40** [Challenging] • Target: Domain and range and continuity • Cross-Lesson: 9.01 Trigonometric Functions

A data set recorded for domain and range and continuity consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

**a.** 17.5

**b.** 18.5

**c.** 19.5 ✓

**d.** 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by $4$ gives $19.5$.

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**41** [Challenging] • Target: Domain and range and continuity • Cross-Lesson: 10.01 Matrices & Systems

For the model $f(x) = 5x + 6$ in domain and range and continuity, evaluate $f(4)$.

**a.** 26 ✓

**b.** 31

**c.** 20

**d.** 30

Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

**42** [Challenging] • Target: Domain and range and continuity • Cross-Lesson: 11.01 Probability & Statistics

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

**a.** $x = 3$ and $x = 9$

**b.** $x = 3$ **and** $x = 8$ ✓

**c.** $x = -3$ and $x = -8$

**d.** $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

**43** [Challenging] • Target: Characteristics from a graph • Cross-Lesson: 2.01 Quadratics & Complex Numbers

A data set recorded for characteristics from a graph consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

**a.** 18.5

**b.** 19.5

**c.** 20.5 ✓

**d.** 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by $4$ gives $20.5$.

**44** [Challenging] • Target: Characteristics from a graph • Cross-Lesson: 3.01 Polynomial Functions

For the model $f(x) = 6x + 7$ in characteristics from a graph, evaluate $f(4)$.

**a.** 31 ✓

**b.** 37

**c.** 24

**d.** 35

Substitute $x = 4$: $f(4) = 6(4) + 7 = 31$.

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**45** [Challenging] • Target: Average rates of change • Cross-Lesson: 4.01 Rational Expressions

For the model $f(x) = 2x + 8$ in average rates of change, evaluate $f(3)$.

**a.** 14 ✓

**b.** 16

**c.** 6

**d.** 18

Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

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